Does the Environmental Kuznets Curve (EKC) Hold for Botswana?

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1 Introduction

In this study, I empirically examined whether the EKC prediction holds for Botswana from 1970 to 2024. To do this, I used a quadratic OLS model to capture the non-linear behavior if it exists. The results were robust, supporting the EKC prediction.

2 The Empirical Model

$$\ln(\mathrm{CO}_2)_t = \beta_0 + \beta_1 \ln(\mathrm{GDP}_c)_t + \beta_2 (\ln(\mathrm{GDP}_c))_t^2 + \varepsilon_t$$

Where:

- $\ln(\mathrm{CO}_2)_t$ is the natural logarithm of carbon dioxide emissions at time t.
- $\ln(\mathrm{GDP}_c)_t$ is the mean-centered natural logarithm of GDP per capita at time t.
- $(\ln(\mathrm{GDP}_c))_t^2$ is the squared term of the mean-centered ln(GDP per capita), capturing the non-linear curvature.

I used the mean-centered versions of GDP per capita to fix multicollinearity issues.

3 Results and Interpretation

I used the World Bank Database from 1970 to 2024 for both variables for statistical analysis.

Variable	Coefficient	Std. Error	t-statistic	p-value	95% CI
Intercept (β_0)	0.7029	0.039	18.231	0.000	[0.626, 0.780]
Log(GDP pc) centered (β_1)	0.6987	0.063	11.037	0.000	[0.572, 0.826]
Log(GDP square) (β_2)	-0.4471	0.082	-5.449	0.000	[-0.612, -0.282]

Table 1: OLS Regression Results for Botswana (1970–2024)

Note: Dependent Variable: $\ln(\mathrm{CO}_2)$. $N = 55$, $R^2 = 0.900$, Adjusted $R^2 = 0.896$, F-statistic = 234.7 ($p = 9.35 \times 10^{-27}$).

$$\ln(\mathrm{CO}_2)_t = 0.7029 + 0.6987 \cdot \ln(\mathrm{GDP}_c)_t - 0.4471 \cdot (\ln(\mathrm{GDP}_c))_t^2$$

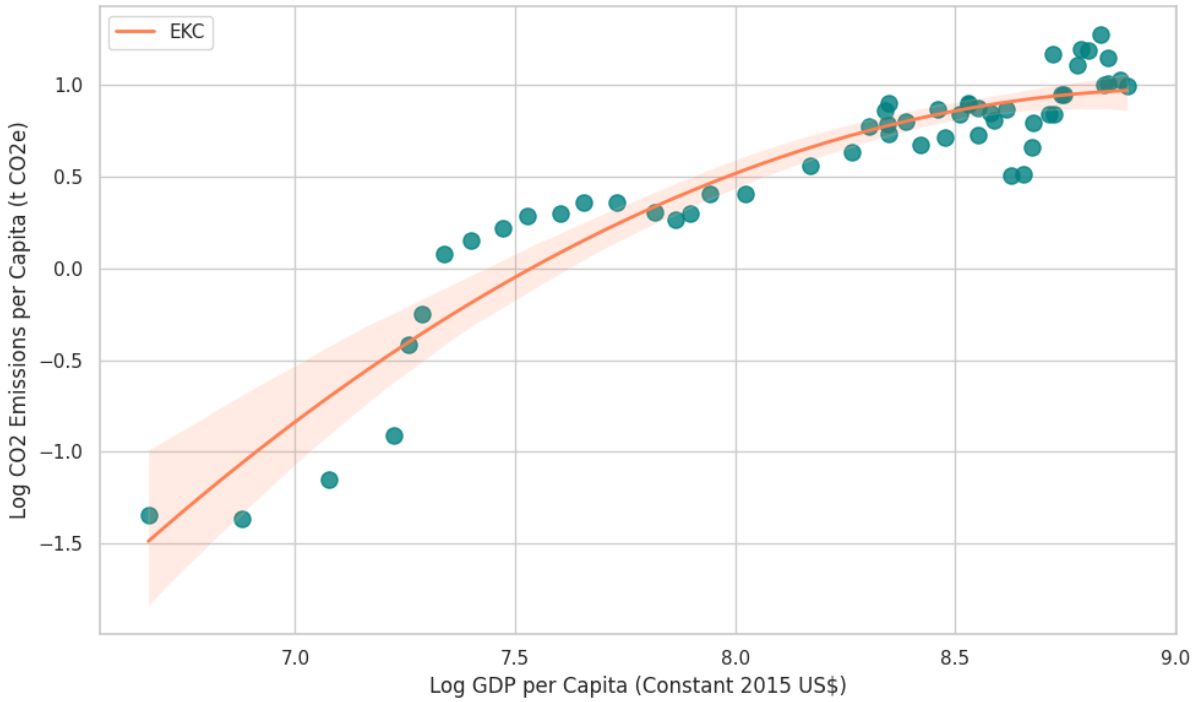


Figure 1: Environmental Kuznets Curve for Botswana (1970–2024)

An adjusted R^2 of 0.896 indicates that approximately 89.6% of the variation in logged CO_2 emissions is explained by GDP per capita and an F-statistic of 234.7 ($p < 0.001$) indicates that the variation of the carbon emissions explained by the model is vastly greater than the unexplained random error. So the model is a good fit.

The coefficients for the linear ($\beta_1 = 0.6987$) and quadratic ($\beta_2 = -0.4471$) terms of log GDP per capita are statistically significant ($p < 0.001$). The positive linear coefficient tells us that the initial economic growth is correlated with higher carbon emissions. However, the negative quadratic coefficient which represents non-linearity creates a bending curve as the country gets richer, creating a negative effect that balances the linear effect. This implies that once Botswana's economy surpasses a critical income threshold, the relationship decouples, leading to a relative reduction in per capita carbon emissions. This is in line with the EKC prediction, so EKC holds well for Botswana.

4 Resources

You can access the code for the regression from here:

https://github.com/fruich/EKC-Regression-for-Botswana